

Statistics, Counting & Probability

ANSWER KEY



The fundamental counting principle

- 1 A restaurant menu has 4 appetizers, 6 main courses, and 3 desserts. How many different 3-course meals are possible?

By the fundamental counting principle: $4 \times 6 \times 3 = 72$.

- 2 A license plate consists of 3 letters followed by 4 digits. How many different license plates are possible, assuming repetition is allowed?

$$26^3 \times 10^4 = 17576 \times 10000 = 175,760,000.$$

Permutations

- 3 In how many ways can 5 different books be arranged on a shelf?

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120.$$

- 4 In how many ways can a president, vice president, and secretary be chosen from a club of 10 members?

$$P(10, 3) = \frac{10!}{7!} = 10 \times 9 \times 8 = 720.$$

Combinations

- 5 In how many ways can a hand of 5 cards be dealt from a standard 52-card deck?

$$\binom{52}{5} = \frac{52!}{5! 47!} = 2,598,960.$$

- 6 A committee of 4 is chosen from 6 men and 5 women. In how many ways can the committee consist of exactly 2 men and 2 women?

$$\binom{6}{2} \times \binom{5}{2} = 15 \times 10 = 150.$$

Probability with combinations

- 7 A bag contains 5 red and 7 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red.

$$P(\text{all red}) = \frac{\binom{5}{3}}{\binom{12}{3}} = \frac{10}{220} = \frac{1}{22}.$$

The binomial probability formula

- 8 A fair coin is flipped 5 times. Find the probability of getting exactly 3 heads, using the binomial probability formula $P(k) = \binom{n}{k} p^k (1-p)^{n-k}$.

$$P(3) = \binom{5}{3} (0.5)^3 (0.5)^2 = 10(0.125)(0.25) = 10(0.03125) = 0.3125.$$

Expected value and variance

- 9 A game gives a payout of 10 dollars with probability 0.3, 5 dollars with probability 0.4, and 0 dollars with probability 0.3. Find the expected payout.

$$\text{Expected value} = 10(0.3) + 5(0.4) + 0(0.3) = 3 + 2 + 0 = 5 \text{ dollars.}$$

Normal distribution and z-scores

- 10 A data set is normally distributed with mean 70 and standard deviation 8. Find the z-score for a data value of 86.

$$z = \frac{86 - 70}{8} = \frac{16}{8} = 2.$$

- 11 Using the empirical rule (68-95-99.7 rule), estimate the percentage of data within 2 standard deviations of the mean.

By the empirical rule, approximately 95% of data in a normal distribution falls within 2 standard deviations of the mean.

A multi-part probability and counting problem

- 12 This question has two parts. A class has 8 boys and 7 girls. A group of 4 students is selected at random to represent the class. (a) Find the total number of ways to select the group of 4 from the 15 students. (b) Find the probability that the group consists of exactly 2 boys and 2 girls.

$$\begin{aligned} \text{(a) } \binom{15}{4} &= \frac{15!}{4! 11!} = 1365. \text{ (b) Ways to choose 2 boys and 2 girls: } \binom{8}{2} \times \binom{7}{2} = 28 \times 21 = 588. \\ \text{Probability: } &\frac{588}{1365} \approx 0.431, \text{ or about } 43.1\%. \end{aligned}$$

Permutations with repeated elements

- 13 Find the number of distinct arrangements of the letters in the word "BANANA".

The word has 6 letters with repeats: A appears 3 times, N appears 2 times. Number of distinct arrangements: $\frac{6!}{3! 2!} = \frac{720}{12} = 60$.

Conditional probability with counting

- 14** A committee of 3 is selected from 5 men and 4 women. Given that at least one woman is selected, find the probability that exactly 2 women are selected.

Total ways with at least 1 woman: total ways $\binom{9}{3} = 84$ minus all-men ways $\binom{5}{3} = 10$, giving $84 - 10 = 74$. Ways with exactly 2 women: $\binom{4}{2} \times \binom{5}{1} = 6 \times 5 = 30$. Conditional probability: $\frac{30}{74} = \frac{15}{37} \approx 0.405$.

Percentiles and standardized scores

- 15** A student scores at the 85th percentile on a standardized test. Explain what this means in terms of the rest of the test-taking population.

Scoring at the 85th percentile means the student performed better than approximately 85% of all test-takers, and approximately 15% of test-takers scored higher than this student.

Mutually exclusive vs. overlapping events

- 16** A die is rolled once. Find $P(\text{rolling an even number or a 5})$, explaining why the addition rule simplifies since these events are mutually exclusive.

Even numbers: 2, 4, 6 (3 outcomes). Since 5 is odd, it cannot overlap with the even outcomes — the events are mutually exclusive. $P(\text{even or 5}) = P(\text{even}) + P(5) = \frac{3}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$.

A multi-part counting and probability problem with restrictions

- 17** This question has three parts. A password consists of 4 characters chosen from the 10 digits 0-9, and repetition is not allowed. (a) Find the total number of possible passwords. (b) Find the number of passwords where the first character is not 0 (a common restriction to avoid leading zeros). (c) Find the probability that a randomly generated valid password (from part b) starts with an even digit.

(a) $P(10, 4) = 10 \times 9 \times 8 \times 7 = 5040$. (b) First character has 9 choices (1-9, excluding 0); remaining 3 characters chosen from the remaining 9 digits without repetition: $9 \times 9 \times 8 \times 7 = 4536$. (c) Even first digits from 1-9: 2, 4, 6, 8 (4 choices). Passwords starting with an even digit: $4 \times 9 \times 8 \times 7 = 2016$. Probability: $\frac{2016}{4536} = \frac{4}{9} \approx 0.444$.

Multiple choice — easy

- 18** A store offers 8 toppings, 2 drink flavors, and 5 wallpaper patterns. How many different combinations of one item from each category are possible?

a. 88
b. 80 ✓
c. 18
d. 21

By the fundamental counting principle: $8 \times 2 \times 5 = 80$.

- 19** A store offers 2 crust types, 7 pant styles, and 3 wallpaper patterns. How many different combinations of one item from each category are possible?

a. 42 ✓
b. 17
c. 23
d. 44

By the fundamental counting principle: $2 \times 7 \times 3 = 42$.

- 20** A store offers 9 font styles, 2 shoe types, and 9 crust types. How many different combinations of one item from each category are possible?

a. 162 ✓
b. 165
c. 171
d. 27

By the fundamental counting principle: $9 \times 2 \times 9 = 162$.

- 21** A store offers 3 drink flavors, 2 appetizers, and 2 font styles. How many different combinations of one item from each category are possible?

a. 12 ✓

b. 7

c. 15

d. 8

By the fundamental counting principle: $3 \times 2 \times 2 = 12$.

- 22** A store offers 6 book genres, 6 appetizers, and 4 shoe types. How many different combinations of one item from each category are possible?

a. 150

b. 30

c. 40

d. 144 ✓

By the fundamental counting principle: $6 \times 6 \times 4 = 144$.

- 23** A store offers 9 main courses, 6 wallpaper patterns, and 7 book genres. How many different combinations of one item from each category are possible?

a. 61

b. 387

c. 378 ✓

d. 51

By the fundamental counting principle: $9 \times 6 \times 7 = 378$.

- 24** A store offers 6 book genres, 4 toppings, and 6 shirt colors. How many different combinations of one item from each category are possible?

a. 144 ✓

b. 150

c. 30

d. 147

By the fundamental counting principle: $6 \times 4 \times 6 = 144$.

- 25** A store offers 5 wallpaper patterns, 5 drink flavors, and 5 book genres. How many different combinations of one item from each category are possible?

a. 30

b. 128

c. 130

d. 125 ✓

By the fundamental counting principle: $5 \times 5 \times 5 = 125$.

- 26** A store offers 7 shoe types, 7 shirt colors, and 2 desserts. How many different combinations of one item from each category are possible?

a. 51

b. 98 ✓

c. 21

d. 105

By the fundamental counting principle: $7 \times 7 \times 2 = 98$.

- 27** A store offers 4 book genres, 2 main courses, and 6 wallpaper patterns. How many different combinations of one item from each category are possible?
- a. 52
 - b. 16
 - c. 48 ✓
 - d. 14

By the fundamental counting principle: $4 \times 2 \times 6 = 48$.

- 28** A store offers 4 main courses, 2 desserts, and 8 toppings. How many different combinations of one item from each category are possible?
- a. 68
 - b. 16
 - c. 20
 - d. 64 ✓

By the fundamental counting principle: $4 \times 2 \times 8 = 64$.

- 29** A store offers 7 crust types, 3 toppings, and 7 appetizers. How many different combinations of one item from each category are possible?
- a. 150
 - b. 28
 - c. 154
 - d. 147 ✓

By the fundamental counting principle: $7 \times 3 \times 7 = 147$.

- 30 A store offers 4 shoe types, 2 desserts, and 5 pant styles. How many different combinations of one item from each category are possible?

a. 44
b. 13
c. 40 ✓
d. 14

By the fundamental counting principle: $4 \times 2 \times 5 = 40$.

- 31 A store offers 6 desserts, 6 main courses, and 9 pant styles. How many different combinations of one item from each category are possible?

a. 60
b. 324 ✓
c. 330
d. 45

By the fundamental counting principle: $6 \times 6 \times 9 = 324$.

- 32 A store offers 7 crust types, 2 toppings, and 5 appetizers. How many different combinations of one item from each category are possible?

a. 77
b. 70 ✓
c. 19
d. 17

By the fundamental counting principle: $7 \times 2 \times 5 = 70$.

Multiple choice — medium

33 In how many ways can a committee of 5 people be chosen from a group of 15 people?

a. 3003 ✓

b. 3008

c. 360360

d. 5005

$$\binom{15}{5} = \frac{15!}{5!(10)!} = 3003.$$

34 In how many ways can a committee of 3 people be chosen from a group of 16 people?

a. 3360

b. 560 ✓

c. 1820

d. 563

$$\binom{16}{3} = \frac{16!}{3!(13)!} = 560.$$

35 In how many ways can a committee of 5 people be chosen from a group of 10 people?

a. 30240

b. 252 ✓

c. 257

d. 210

$$\binom{10}{5} = \frac{10!}{5!(5)!} = 252.$$

36 In how many ways can a committee of 3 people be chosen from a group of 12 people?

a. 1320

b. 220 ✓

c. 223

d. 495

$$\binom{12}{3} = \frac{12!}{3!(9)!} = 220.$$

37 In how many ways can a committee of 8 people be chosen from a group of 18 people?

a. 43766

b. 1764322560

c. 48620

d. 43758 ✓

$$\binom{18}{8} = \frac{18!}{8!(10)!} = 43758.$$

38 In how many ways can a committee of 2 people be chosen from a group of 12 people?

a. 220

b. 66 ✓

c. 68

d. 132

$$\binom{12}{2} = \frac{12!}{2!(10)!} = 66.$$

39 In how many ways can a committee of 12 people be chosen from a group of 15 people?

a. 105

b. 455 ✓

c. 467

d. 217945728000

$$\binom{15}{12} = \frac{15!}{12!(3)!} = 455.$$

40 In how many ways can a committee of 6 people be chosen from a group of 15 people?

a. 5011

b. 6435

c. 5005 ✓

d. 3603600

$$\binom{15}{6} = \frac{15!}{6!(9)!} = 5005.$$

41 In how many ways can a committee of 8 people be chosen from a group of 14 people?

a. 3011

b. 3003 ✓

c. 2002

d. 121080960

$$\binom{14}{8} = \frac{14!}{8!(6)!} = 3003.$$

42 In how many ways can a committee of 5 people be chosen from a group of 8 people?

a. 56 ✓

b. 61

c. 6720

d. 28

$$\binom{8}{5} = \frac{8!}{5!(3)!} = 56.$$

43 In how many ways can a committee of 3 people be chosen from a group of 13 people?

a. 286 ✓

b. 289

c. 715

d. 1716

$$\binom{13}{3} = \frac{13!}{3!(10)!} = 286.$$

44 In how many ways can a committee of 3 people be chosen from a group of 8 people?

a. 56 ✓

b. 59

c. 336

d. 70

$$\binom{8}{3} = \frac{8!}{3!(5)!} = 56.$$

45 In how many ways can a committee of 6 people be chosen from a group of 8 people?

a. 8

b. 28 ✓

c. 34

d. 20160

$$\binom{8}{6} = \frac{8!}{6!(2)!} = 28.$$

46 In how many ways can a committee of 3 people be chosen from a group of 15 people?

a. 1365

b. 2730

c. 458

d. 455 ✓

$$\binom{15}{3} = \frac{15!}{3!(12)!} = 455.$$

47 In how many ways can a committee of 4 people be chosen from a group of 9 people?

a. 130

b. 126 ✓

c. 3024

d. 135

$$\binom{9}{4} = \frac{9!}{4!(5)!} = 126.$$

Multiple choice — hard

48 A bag contains 7 red and 6 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red.

a. 10/143

b. 7/13

c. 35/286 ✓

d. 35/13

$$P(\text{all red}) = \frac{\binom{7}{3}}{\binom{13}{3}} = \frac{35}{286}.$$

49 A bag contains 4 red and 5 blue marbles. If 4 marbles are drawn at random without replacement, find the probability all 4 are red.

a. 5/126

b. 4/9

c. 1/126 ✓

d. 1/9

$$P(\text{all red}) = \frac{\binom{4}{4}}{\binom{9}{4}} = \frac{1}{126}.$$

- 50 A bag contains 5 red and 5 blue marbles. If 4 marbles are drawn at random without replacement, find the probability all 4 are red.

a. $1/14$

b. $1/2$

c. $1/21$

d. $1/42$ ✓

$$P(\text{all red}) = \frac{\binom{5}{4}}{\binom{10}{4}} = \frac{5}{210} = \frac{1}{42}.$$

- 51 A bag contains 8 red and 5 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red.

a. $56/13$

b. $28/143$ ✓

c. $8/13$

d. $5/143$

$$P(\text{all red}) = \frac{\binom{8}{3}}{\binom{13}{3}} = \frac{56}{286} = \frac{28}{143}.$$

- 52 A bag contains 6 red and 6 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red.

a. $1/11$ ✓

b. $5/3$

c. $2/11$

d. $1/2$

$$P(\text{all red}) = \frac{\binom{6}{3}}{\binom{12}{3}} = \frac{20}{220} = \frac{1}{11}.$$

- 53 A fair coin is flipped 4 times. Find the probability of getting exactly 2 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.

- a. $1/2$
- b. $3/8$ ✓
- c. $3/4$
- d. $5/8$

$$P(2) = \binom{4}{2}(0.5)^4 = 6 \times \frac{1}{16} = \frac{6}{16} = \frac{3}{8}.$$

- 54 A fair coin is flipped 8 times. Find the probability of getting exactly 5 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.

- a. $5/16$
- b. $1/4$
- c. $7/32$ ✓
- d. $9/32$

$$P(5) = \binom{8}{5}(0.5)^8 = 56 \times \frac{1}{256} = \frac{56}{256} = \frac{7}{32}.$$

- 55 A fair coin is flipped 4 times. Find the probability of getting exactly 1 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.

- a. 1
- b. $1/4$ ✓
- c. $3/4$
- d. $1/2$

$$P(1) = \binom{4}{1}(0.5)^4 = 4 \times \frac{1}{16} = \frac{4}{16} = \frac{1}{4}.$$

- 56 A fair coin is flipped 7 times. Find the probability of getting exactly 1 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.

- a. 1/16
- b. 1
- c. 9/128
- d. 7/128 ✓

$$P(1) = \binom{7}{1}(0.5)^7 = 7 \times \frac{1}{128} = \frac{7}{128}.$$

- 57 A fair coin is flipped 5 times. Find the probability of getting exactly 4 heads, using the binomial probability formula $P(k) = \binom{n}{k}p^k(1-p)^{n-k}$.

- a. 1
- b. 5/32 ✓
- c. 7/32
- d. 3/16

$$P(4) = \binom{5}{4}(0.5)^5 = 5 \times \frac{1}{32} = \frac{5}{32}.$$

- 58 Find the number of distinct arrangements of the letters in the word "PIZZA".

- a. 60 ✓
- b. 120
- c. 58
- d. 63

The word has 5 letters with repeats: Z appears 2 times. Number of distinct arrangements:
 $5!/(2!) = 120/2 = 60.$

59 Find the number of distinct arrangements of the letters in the word "APPLE".

a. 60 ✓

b. 58

c. 63

d. 120

The word has 5 letters with repeats: P appears 2 times. Number of distinct arrangements:
 $5! / (2!) = 120/2 = 60$.

60 Find the number of distinct arrangements of the letters in the word "TENNESSEE".

a. 3780 ✓

b. 362880

c. 3684

d. 7560

The word has 9 letters with repeats: E appears 4 times, N appears 2 times, S appears 2 times. Number of distinct arrangements: $9! / (4! 2! 2!) = 362880/96 = 3780$.

61 Find the number of distinct arrangements of the letters in the word "COMMITTEE".

a. 45360 ✓

b. 90720

c. 45352

d. 362880

The word has 9 letters with repeats: E appears 2 times, M appears 2 times, T appears 2 times. Number of distinct arrangements: $9! / (2! 2! 2!) = 362880/8 = 45360$.

62 Find the number of distinct arrangements of the letters in the word "BANANA".

a. 120

b. 48

c. 720

d. 60 ✓

The word has 6 letters with repeats: A appears 3 times, N appears 2 times. Number of distinct arrangements: $6! / (3! 2!) = 720/12 = 60$.